Low-Loss Waveguide Bends by Advanced Shape for Photonic Integrated Circuits

Jeong Hwan Song, Tangla D. Kongnyuy, Peter De Heyn, Sebastien Lardenois, Roelof Jansen, and Xavier Rottenberg

***Abstract*—We report the design and experiment of advanced bends which are composed of widening width and varying radius. Advanced bends are based on silicon nitride waveguide for the wavelength of 850 nm and silicon waveguide for the wavelength of 1550 nm. The reduction of bending loss from 0.13 to 0.04 dB/90° by the advanced shape on a silicon nitride waveguide bend with 15 μm-radius at the wavelength of 850 nm was achieved. The bending loss from the silicon advanced bend was measured to be 0.012 dB/90° when $R = 2$ μm at the wavelength of 1550 nm. This is the best result of 2 μm-bends in a 220 nm-SOI platform at 1550 nm.**

***Index Terms*—Bends, silicon, silicon nitride, SOI, waveguides.**

I. Introduction

SILICON and silicon nitride photonic integrated circuits (PICs) can be fabricated by the compatible process of a complementary metal-oxide-semiconductor line [1], [2] for the many applications such as life science components, sensors [3]–[5], telecoms and datacoms [6]. Silicon and silicon nitride PICs have many advantages. Among them, the key advantage is that devices can be miniaturized. The compactness of devices by silicon and silicon nitride waveguides can be achieved by making low loss bends with high index contrasts. The index contrasts are ~45% and ~20% defined as $\Delta = ({n_{\rm core}}^2 - {n_{\rm clad}}^2)/(2{n_{\rm core}}^2)$ for silicon and silicon nitride waveguides, respectively. Here, $n_{\rm core}$ *and* $n_{\rm clad}$ are refractive indices of core and cladding materials, respectively. In the case of silicon waveguides, the index contrast of silicon waveguide is much higher than that of silicon nitrides, however, the bending radius of smaller than 5 μm should not be recommended for the practical use. Because the loss from the bends having below the radius of 5 μm would not be ignorable at complexed PICs. Recently, PICs are being denser, and some complex components require several tens or hundreds of bends such as switch fabrics [7], multi-channel lidar applications [8]. In this case, the bending loss will be problematic. Therefore, reducing bending loss is essential for high-performance PICs tiny footprint required. There are several methods proposed to reduce bending loss. A widening bend with offsets [9], [10] and Euler (or clothoid) bend structures [11]–[15] are reported for reducing a bending loss for a small bending radius. In the case of widening bends with offsets [9], modal confinements can be increased on the wider waveguide and consequently, the radiation losses are decreased. Additionally, the offset can be decreased to the junction loss between straight and bending waveguides. For the replacement of the offset in order to decrease the junction loss, Euler bends can be used. The junction loss, which comes from the mismatch of shape and alignment [9] between straight and bending waveguide modes, is a big portion of total loss in a bend connected to a straight waveguide. Another method to reduce bend losses is by using a multimode waveguide composed by contour trajectories. A U-shape of 180° bend was reported with the method [16]. Song, *et al.* recently reported bends designed by the combination of widening width and varying radius (like Euler bend), called as *advanced bends* [17]. Fig. 1 shows the shape of the bend and parameters. The widening waveguide can allow a highly confined mode in a waveguide. Simultaneously, the varying radius structure can minimize the junction loss, which is a similar effect of waveguide offset between the straight and bending waveguides.

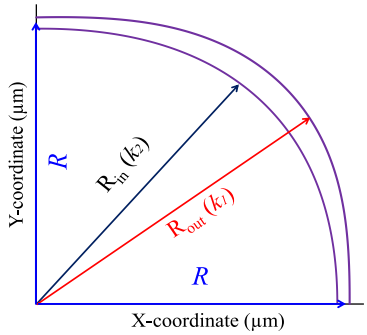


Fig. 1. Advanced bend structure and its design parameters.

In this paper, we demonstrate the performance of the advanced bends on silicon and silicon nitride waveguides at the wavelengths of 1550 nm and 850 nm, respectively.

II. Design Approaches

The modal confinement and effective index of the bending waveguides are weaker than those of straight waveguides when both waveguides have the same dimension. The radiation loss

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The authors are with the IMEC, 3001 Leuven, Belgium (e-mail: jeonghwan.song@imec.be; Tangla.David.Kongnyuy@imec.be; Peter.DeHeyn@imec.be; Sebastien.Lardenois@imec.be; Roelof.Jansen@imec.be; Xavier.Rottenberg@imec.be).

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by bending can be reduced by changing the dimension of the waveguides [9]. The height of waveguides can be determined by an initial design of the height or a commercial SOI (silicon on insulator) wafer used. Varying the height of the waveguides is not easy. But varying the width of waveguides can be an easy solution to change waveguide dimension to increase the modal confinement and effective index on bending waveguides. Not to generate or not to be affected by multi-modes, the width of bends must be gradually increased and decreased as tapering bends. This tapering bend will work as a mode filter. Varying radius like Euler bends helps to reduce transition loss between the bend and straight waveguides. Therefore, the bend shape profile can be made by sinusoidal curves as expressed below [9].

$$R_{out}(\theta_n) = R + \frac{1}{2}\left(\omega_{n-1} + \frac{1}{k_1}\sin(q\theta_n)\right) \quad (1)$$

$$R_{in}(\theta_n) = R - \frac{1}{2}\left(\omega_{n-1} + \frac{1}{k_2}\sin(q\theta_n)\right) \quad (2)$$

Here, $q = 4$ for the full cycle of sine function within 90 degrees and R is the radius. $\theta_n = n \times \Delta\theta$, where we used the value as $\Delta\theta = 1^0$ and $n = 90$. k_1 and k_2 are bending coefficients. ω_{n-1} is the varying width of the waveguide by segments.

The varying radius (R_k) and width (ω_k) as a function of the angle of bends can be described as

$$R_k = \frac{1}{2}\left(R_{out}(\theta_n) + R_{in}(\theta_n)\right) \quad (3)$$

$$\omega_k = R_{out}(\theta_n) - R_{in}(\theta_n) = \omega_{n-1} + \frac{1}{2}\left(\frac{1}{k_1} + \frac{1}{k_2}\right)\sin(q\theta_n). \quad (4)$$

Equations of (1) to (4) can be used for the design of bend shapes universally. The usages of Eqs. (1) to (4) are below.

1) When $k_1 = k_2$, the bent shape will be gradually widening up to 45° and then narrowing up to 90° without changing radius.
2) When k_1 and k_2 are infinite, consequently $R_k = R$ and $\omega_k = \omega_0$. The bend will be a conventional bend.
3) When $k_1 = -k_2$, namely, $\omega_k = \omega_0$ and $R_k \neq R$, the geometry of bends is like the Euler bend. The curvature ($=1/R$) is gradually increased up to 45° and decreased symmetrically without changing waveguide width. The Euler bend is mathematically defined as a bend with the curvature varying linearly along the path length [12]. However, using equations, the curvature of bends gradually varies as sinusoidal curves with equivalent width. So, in this paper, we called it as *equi-width bends*.
4) When $k_1 \neq k_2$, the bent shape will be gradually widening up to 45° and then narrowing up to 90°. The radius is changed. We call it *advanced bends* as shown in Fig. 1. The radius and width are changed simultaneously in the advanced bend.

PICs with silicon or silicon nitride waveguides use mainly transverse electric (TE) mode lights. Therefore, we design waveguide bends only for TE mode in this paper. It can be used for the design of bends for transverse magnetic (TM) mode.

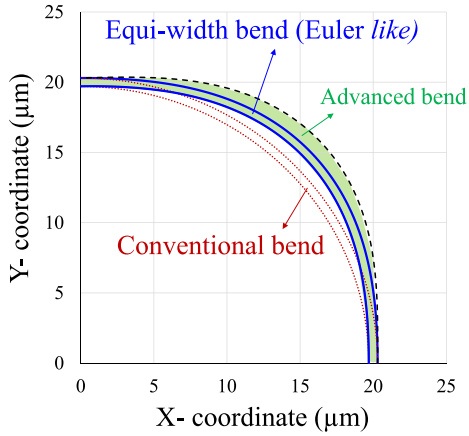


Fig. 2. Shape comparison: Advanced bend vs. conventional bend vs. equi-width bend.

TABLE I
SPECIFICATIONS AND COMMON PARAMETERS OF WAVEGUIDES

Parameter \ material	Silicon nitride	Silicon
Wavelength	850nm	1550nm
Index of core	1.88	3.47
Index of cladding	1.46	1.45
Waveguide height	300nm	220nm
Waveguide width (for single mode)	600nm	450nm

TABLE II
OPTIMIZED VALUES OF K_1 AND K_2 FOR SILICON NITRIDE EQUI-WIDTH BENDS

R	5μm	10μm	15μm	20μm
k_1	90	45	15	10
k_2	−90	−45	−15	−10

Fig. 2 shows the comparisons of shapes of advanced, conventional, and equi-width bends using Eqs. (1)–(4). The specifications of waveguides and common design parameters are summarized in Table I.

III. SILICON NITRIDE ADVANCED BENDS

A. Design of Bends for 850 nm

Using a 3-dimensional Finite-Difference Time-Domain (3D-FDTD) method, we optimized the advanced bend shapes of silicon nitride waveguide when R = 5, 10, 15, and 20 μm at the wavelength of 850 nm. The waveguide dimension is a width of 600 nm by a height of 300 nm. Refractive indices of core and cladding layers are 1.88 and 1.46 at the 850 nm-wavelength, respectively. To find optimum values of k_1 and k_2, we swept the values of k_1 and k_2 using 3D-FDTD simulations. The optimum values of k_1 and k_2 are summarized in Table II and III for the equi-width and advanced bends, respectively. The shapes of the bends, varying radii and widths are shown in Fig. 3. Fig. 4 shows the 3D-FDTD simulation results with the optimum values. The

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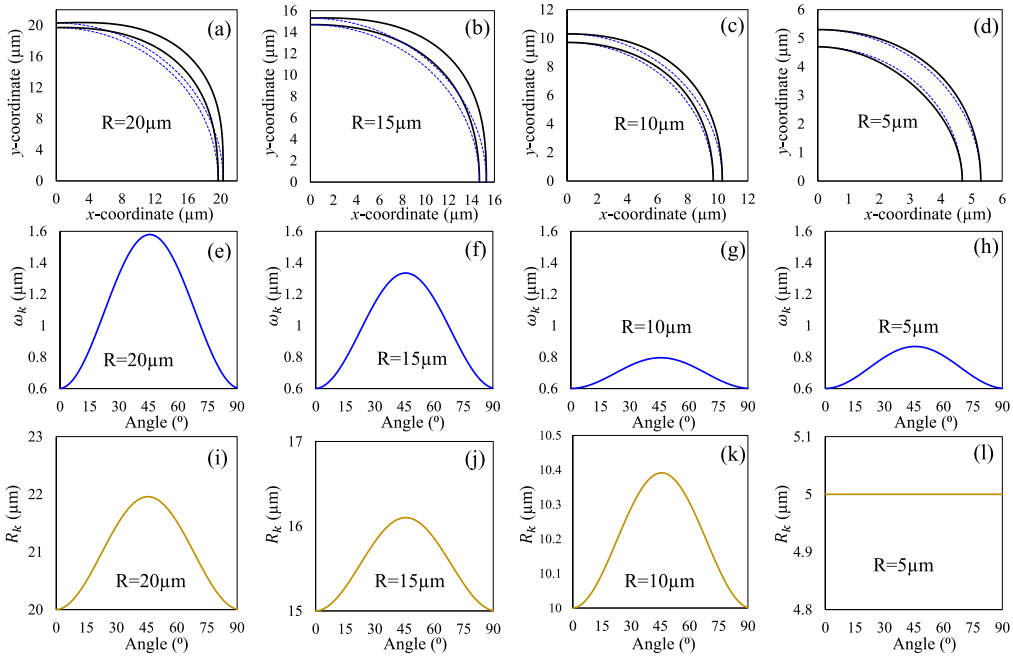


Fig. 3. (a)–(d) shapes of advanced bends with different radii and comparison with normal bends on silicon nitride waveguide bends, (e)–(h) width change as a function of angle on advanced bends, and (i)–(l) radius change as a function of angle.

TABLE III
OPTIMIZED VALUES OF K_1 AND K_2 FOR SILICON NITRIDE ADVANCED BENDS

R	5μm	10μm	15μm	20μm
k_1	110	30	10	6
k_2	110	−50	−20	−10

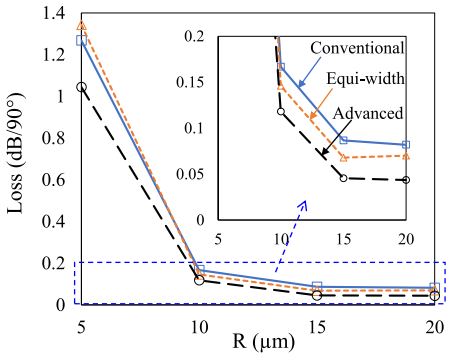


Fig. 4. Simulation results for conventional, equi-width, and advanced bends on a silicon nitride platform with different radii for 850 nm wavelength.

simulations show that the advanced bends have improved performances over equi-width and conventional bends. In the case of the R of 5 μm, the varying radius did not help to decrease bending loss. The equi-width bend also does not contribute to decreasing the bending loss [11]. Only the widening width contributes to decreasing the loss.

B. Experiments

The silicon nitride waveguides and bends were fabricated by PECVD (plasma-enhanced chemical vapor deposition) method.

TABLE IV
SUMMARY OF MEASURED SILICON NITRIDE BENDING LOSSES FOR 850 NM

Bend \ R	5μm	10μm	15μm	20μm
Conventional	1.95dB/90°	0.43dB/90°	0.13 dB/90°	0.06dB/90°
Equi-width	2.52dB/90°	0.33dB/90°	0.08 dB/90°	0.03dB/90°
Advanced	*1.82*dB/90°	*0.25*dB/90°	*0.04 dB/90°*	*0.01*dB/90°

The platform's propagation loss is less than 0.3 dB/cm with an 850 nm light source. The cross-section of platforms includes a silicon substrate, an under cladding layer with a thickness of 3.3 μm, a silicon nitride core layer with a thickness of 0.3 μm, and a top cladding layer with the thickness of 2μm. Four different numbers of bends with the radii of 5, 10, 15 and 20 μm were measured. We tested 1) conventional bends, 2) equi-width bends, and 3) advanced bends for comparison. All test structures have in- and output-grating couplers. A typical grating coupler measurement setup was used for the test of bending losses. The output of an 850 nm laser source was connected to an input-polarization maintaining fiber (PMF) for using TE mode. The output-PMF was connected to a broadband power detector. Each of the input and output was attached to a precision tilting mount on three-axis automatic-translation stages. The angles of the input and output fibers were 10°. Fig. 5 shows the measurement results with the comparison of a conventional, equi-width, and advanced bends. When $R = 5$ μm, the loss of conventional bend is lower than that of the equi-width bend as we simulated. Only the widening of the waveguide helped to decrease the bending loss. The advanced bend has the lowest loss among tested bends. Measured bending losses are summarized in Table IV.

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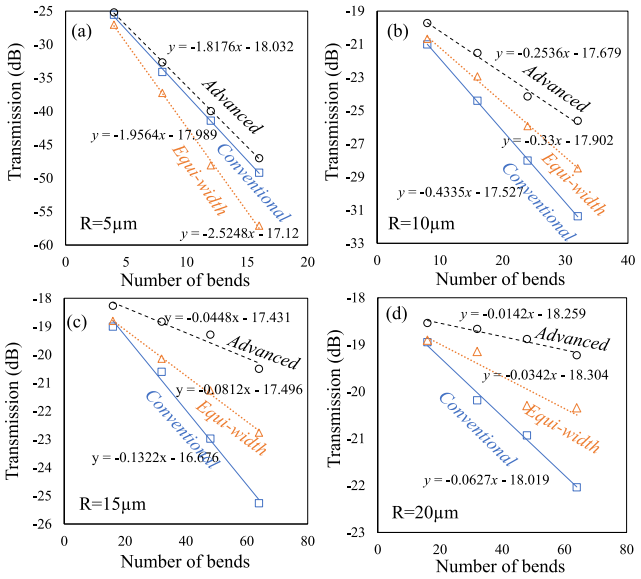


Fig. 5. Measurement results of silicon nitride waveguide bending losses at 850 nm when (a) R = 5 μm, (b) R = 10 μm, (c) R = 15 μm, and (d) R = 20 μm.

TABLE V
SUMMARY OF DESIGN PARAMETERS FOR SILICON WAVEGUIDE BENDS

Parameters / R	1μm	2μm	3μm	Note
R_k	0.951μm	2.122μm	3.356μm	at 45°
ω_k	0.645μm	0.531μm	0.575μm	at 45°
k_1	300 (600)*	90 (120)*	35 (50)*	*equi-width
k_2	100 (−600)*	−180 (−120)*	−50 (−50)*	

IV. SILICON ADVANCED BENDS FOR C-BAND

A. Design and Simulation

Silicon waveguide bends have low bending losses due to the high index contrast [18]. However, for denser PICs with smaller footprint, extremely small radius bending waveguides are necessary. However, the bending radius of smaller than 5 μm should not be used for the practical design. Because the loss from the bends having below the radius of 5 μm is rapidly increased at longer than 1525 nm, near cut-off wavelength [18]. Therefore, we applied this technology to silicon waveguide bends for the wavelength of 1550 nm. To make the optimal shape of advanced bends on the silicon waveguides, we also swept the values of k_1 and k_2 and then implemented 3D-FDTD simulations. For example, the optimized values for k_1 and k_2 range from 90 to 120 and from −190 to −150, respectively, when R = 2 μm as shown in Fig. 6. The optimal parameters we found are summarized in Table V. Fig. 7(a) shows the advanced bend shape with the comparison of a normal bend when R = 2 μm. In this case, varying radius and width as a function of angles are shown in Fig. 7(b). It shows the width of the bend is gradually widening up to 45° and then narrowing again up to 90°. The optimized maximum-width of the bend is 531 nm and the radius of the advanced bend is varied from 2 μm up to 2.12 μm. We also plot the electric field distribution of a normal, equi-width, and advanced

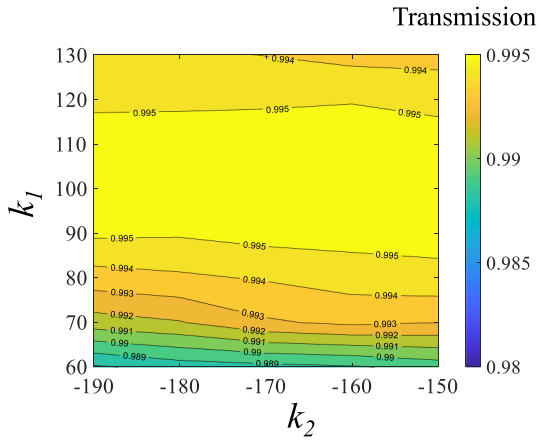


Fig. 6. Optimization of parameters k_1 and k_2 for the silicon advanced bends when $R = 2$ μm.

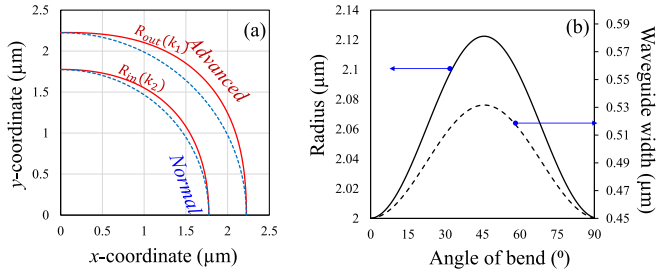


Fig. 7. (a) Bend shape comparison between normal (dotted line) and advanced (solid line) bends and (b) varying radius (solid) and width (dotted) as a function of the angle of bend.

bends in Fig. 8. The radiation losses of the normal (Fig. 8(a)) and equi-width (Fig. 8(b)) bends is more widely distributed in the window than the advanced bend (Fig. 8(c)). The inset of Fig. 9(a) shows the shape of the 1 μm-radius bend structure with the comparison of a conventional bend. Fig. 9 shows simulation results using the optimized k values for the radii of 1, 2, and 3 μm by 3D-FDTD. The results show that advanced bends have better loss characteristics than conventional and equi-width bends. The widen bending waveguide with a changing radius can effectively reduce the loss. At longer wavelength, bending losses increase since the mode approaches cut-off. However, advanced bends relatively have less increment of bending loss due to the widen width.

B. Experiments

The silicon bends were fabricated by IMEC's standard process [19] based on 220 nm-silicon and 2 μm-buried oxide (BOX) silicon on insulator (SOI) wafers. The silicon is patterned using 193 nm dry lithography and reactive ion etching (RIE) to realize reliably features as small as 150 nm. For the grating couplers and test structures, the 220 nm silicon layer is patterned in two levels like the full etch for removing the full silicon thickness in order to make a waveguide core, and the shallow etches of ∼70 nm from the top for grating teeth. The patterned silicon is then clad with High-Density Plasma Chemical Vapor Deposition (HDP-CVD)

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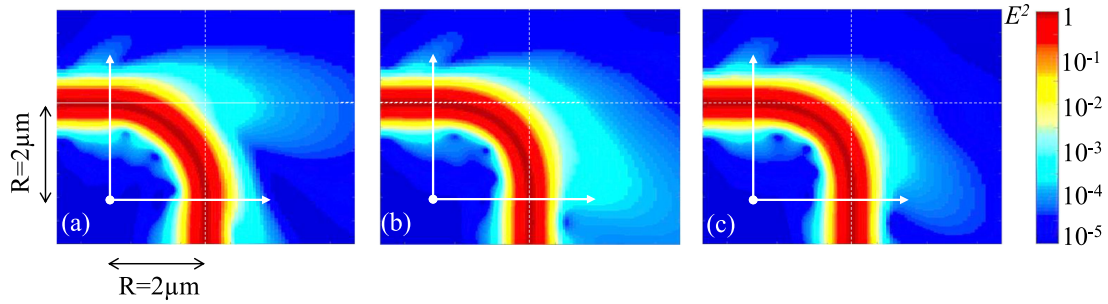


Fig. 8. Electric field plots on silicon bend with a radius of 2 μm. (a) Normal (conventional), (b) equi-width, and (c) advanced bends.

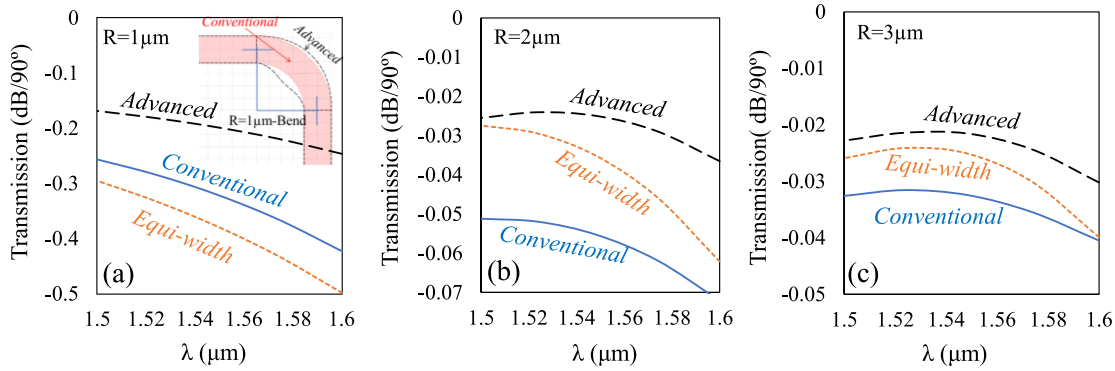


Fig. 9. Simulation results of conventional, equi-width, and advanced bends for silicon waveguides (a) when R = 1 μm, (b) R = 2 μm, and (c) R = 3 μm. The inset of (a) shows the shapes of 1 μm-radius silicon bends.

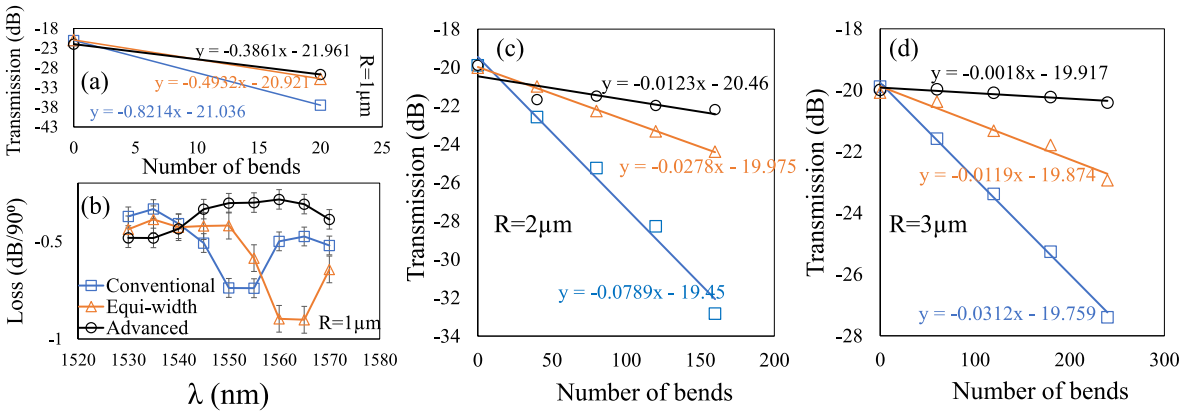


Fig. 10. Measurement data of silicon bends: (a) and (b) R = 1 μm, (c) R = 2 μm, and (d) R = 3 μm. Here, the measurements were implemented at the wavelength of 1550 nm for (a), (c), and (d).

oxide filling narrow trenches such as grating teeth with aspect ratios up to 2:1 and lead to low optical propagation loss. This oxide is planarized with chemo-mechanical polishing (CMP) to obtain a planar surface prior to further processing with the polysilicon module. This extra waveguiding layer of polysilicon allows the realization of higher efficiency grating couplers by increasing the grating directionality. The in-and outputs of all test-devices were polysilicon overlayered grating couplers. The coupling efficiencies of those couplers are over 50%.

Four different numbers of bends with the radii of 1, 2, and 3 μm were fabricated and measured on conventional, equi-width, advanced bends and a reference waveguide. The measurement setup was used with the TE mode of a C-band source. The angles of input and output fibers were 10°. Fig. 10 shows the measurement results of all bends. As we simulated, the loss characteristics of advanced bends are better than the others. The bending loss is dramatically decreased by widening and varying radius. For the case of 1 μm-bends, the results of conventional and equi-width bends show that dips are observed at near 1.55 and 1.56 μm as similarly shown in [18]. We measured only one test set which contains 20 bends. We could not measure other sets of test bends because of too many losses. However, we measured it several times to convince our measurements. The source of dips or resonance observed in the spectrum is not clear. However,

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TABLE VI
SUMMARY OF PUBLISHED WORKS FOR SMALLER THAN 5 μM BENDING RADIUS (COMPARISON OF EXPERIMENTAL BENDING LOSSES ONLY)

Reference	Working group	Year	Radius (μm)	Height (nm)	Width (nm)	Methods	Wavelength (nm)	Loss (dB/90°)	Note
This work	IMEC	2019	1*	220	645**	Advanced	1550	0.35±0.05	*Effective R
			2*	220	531**	Advanced	1550	0.012	**Max width
			3*	220	575**	Advanced	1550	0.0018	
[11]	Hokkaido Univ.	2017	4*	210	400	Euler	1550	0.002	*Effective R
[12]#	VTT	2013	1.27*	4000	900	Euler	1550	0.09** (TE) 0.17** (TM)	*Effective R ***dB/180°* # U-bend
[13]	U of Ghent-Imec	2011	3	220	460	Spline (10°)	1550	0.009	
[16]#	U of Karlsruhe etc.	2007	1.5*	340	1200**	Contour trajectories	1550	0.35***	*Effective R **Max width ****dB/180°* # U-bend
[18]	IBM	2004	1	220	445	Normal	***1500***	0.086±0.005	0.2-0.3dB/90° at >1525nm
			2	220	445	Normal	***1500***	0.013±0.005	
			5	220	445	Normal	***1500***	0±0.005	

TABLE VII
SUMMARY OF MEASURED SILICON WAVEGUIDE BENDING LOSSES AT 1550 NM

Bend / R	1μm	2μm	3μm
Conventional	0.82dB/90°	0.078dB/90°	0.0312 dB/90°
Equi-width	0.49dB/90°	0.027dB/90°	0.0119 dB/90°
Advanced	*0.38*dB/90°	*0.012*dB/90°	*0.0018dB/90°*

for the case of advanced bends, the dip was not seen. Also, we did not observe the dips or resonances when measuring all bends of $R = 2$ and 3 μm with a wide range of C-band. Table VI summarized our results and other's published experimental results. The advanced bends have good bending-loss performances as shown in the comparison. Table VII summarized measurement results.

V. DISCUSSIONS

For further improvement, we added the offsets [10] to silicon nitride bends operational at 850 nm experimentally. We believe the offset helps to reduce losses when using the advanced or equi-width (Euler-like) bend for smaller bending radii. The optimal offset of the advanced bend with a tiny bending radius is smaller than that of a conventional bend. Fig. 11 shows the measurement results for test structures with offsets. We applied offsets of 60 nm and 20 nm on the bends for R equal to 5 μm and 10 μm, respectively. The advanced bends with offsets show an improvement compared to the bends without offsets as shown in Fig. 11. When R is 10 μm, the junction mismatch loss is more critical than the radiation loss. Therefore, adding an offset or tuning the radius is more effective to reduce bending losses than increasing the width. All measured values are shown in Fig. 11. Fig. 12 shows the simulation results of silicon advanced bends for $R = 2$ μm. The optimal 10 nm-offset reduces bending losses for the advanced bend with a radius of 2 μm. The optimal offset

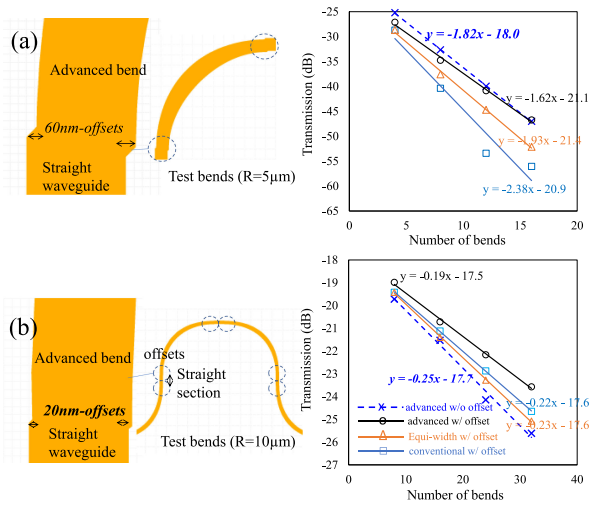


Fig. 11. Experimental study of offset effect on bends: Silicon nitride waveguide bends with offsets (a) when $R = 5$ μm and (b) when $R = 10$ μm. The fitting formulas indicate measured bending loss. The fitting formula in the italic font in blue represents the loss data from advanced bend without the offset.

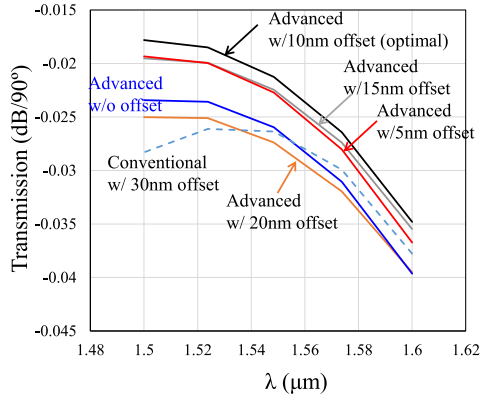


Fig. 12. Theoretical study of offset effect on silicon advanced bends when $R = 2$ μm.

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for a normal bend is found to be 30 nm, which is larger than that of the advanced bend as we mentioned.

VI. Conclusion

The bends based on silicon nitride for 850 nm and silicon for 1550 nm have been experimentally demonstrated. The reduction of bending loss was from 0.13 to 0.04 dB/90° by the advanced shape on a silicon nitride waveguide bend when $R = 20$ μm at 850 nm. For the case of silicon bends, the loss from the advanced bend was measured to be 0.012 dB/90° when $R = 2$ μm at 1550 nm. This is the best result of 2 μm-bends in a 220 nm-SOI platform at 1550 nm. Widening width can increase the confinement of mode and effective index of a bent waveguide. As a result of widening, radiation loss in the bend is decreased. Varying radius can decrease the mismatch loss between straight and bent waveguides. We also discussed the optimal offset on the advanced bends for further improvement. This technology will work universally on photonic waveguide bends with all range of wavelengths.

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Jeong Hwan Song received the B.Sc. degree in physics in 1997 and the M.Sc. and Ph.D. degrees in electrical and computer engineering in 2000 and 2006, respectively. Since 2014, he has been working with the IMEC, Leuven, Belgium. Prior to joining IMEC, he was with Samsung Electronics (Korea), Tyndall National Institute (Ireland), IME A∗STAR (Singapore), and IBM T. J. Watson Research Center (USA) from 2000 to 2014. He served the Korean Army from January 1992 to April 1994. His research interests include integrated photonics and optical components.

Tangla D. Kongnyuy was born in Jakiri, West Cameroon, in 1986. He received the B.Sc. degree in physics from the Anglo-Saxon University of Buea, Buea, Cameroon, in 2011, and the M.Sc. degree in photonics from Ghent University, Ghent, Belgium, in 2014. In 2014, he started TANGLATEC, an engineering consulting company in Ghent. He is currently working on optical characterization and automation of advanced measurement systems that are used for the characterization and validation of photonic integrated circuits (PICs) at IMEC.

Peter De Heyn received the master's degree in photonics engineering from Ghent University, Ghent, Belgium and Vrije Universiteit Brussel, Brussel, Belgium in 2009 and the Ph.D. degree from Ghent University in 2014. He has been a R&D Engineer with IMEC, Leuven, Belgium, since 2014 and is specialized in silicon photonics design, modeling, measurements, and analysis.

Sebastien Lardenois, biography not available at the time of publication.

Roelof Jansen received a M.Sc. degree in electronic engineering and computer science from the University of Stellenbosch, Stellenbosch, South Africa, in 2008. He joined IMEC, Leuven, Belgium, in 2007, where he is currently the Team Leader for the Photonic Microsystems team, part of the Wave-based Sensing and Actuation (WSA) group. The team specializes in systems based on integrated photonic platforms such as optical beamformers, spectrometers, optical sensors, flat optics, and more. He has more than 14 years of experience in the fields of Photonics, RF, and MEMS and has (co-)authored more than 50 peer reviewed publications and several patents.

Xavier Rottenberg received the M.Sc. degree in physics engineering and the DEA in theoretical physics from the Université Libre de Bruxelles, City of Brussels, Belgium, in 1998 and 1999, respectively and the Ph.D. degree in electrical engineering from KU Leuven, Leuven, Belgium, in 2008. He worked for one year at the Royal Meteorological Institute of Belgium in the field of remote sensing from Space. Since 2000, he has been with IMEC Leuven, where he contributes to research in the field of RF, RF-MEMS, photonics, and microsystems modeling/integration. He currently leads the Wave-based Sensing and Actuation (WSA) Group, working among other topics on photonics, sensors, acoustics, and M/NEMS. He has (co-)authored more than 120 peer reviewed publications and has been issued various patents. He was a Lecturer or Invited Speaker with DTIP (‘05), EuroSimE (‘08), ESREF (‘08), ESoA (‘10/‘12/‘14), RF-MEMS summer school (‘13), FETCH 2016, NERF Neurotechnology Symposium (‘16), OpenMinds (‘17), and Nanoworkshop at Beijing University (‘18).

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